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Clustering U.S. States by Chronic Disease Indicators

Introduction

Chronic diseases such as cancer, diabetes, and alcohol-related health issues are major public health concerns in the United States and have impacted many families. These conditions vary significantly across different states and populations, influenced by socioeconomic, geographic, and demographic factors. Understanding regional,cultural, economic patterns are crucial for us to understand what can be done to help to prevent these Chronic diseases.

My thesis explores how chronic disease burdens vary across U.S. states using stratified health data collected by the CDC. I hypothesize that clustering states by health indicators will reveal distinct public health profiles.

Using the U.S. Chronic Disease Indicators dataset from Data.gov, I conducted data cleaning, encoding, and unsupervised learning via KMeans clustering. I aggregated the data across topics and demographic stratifications to produce a health profile for each state.

The results revealed four distinct clusters of states, with one cluster containing a clear outlier with extremely elevated alcohol-related health indicators.

Methods

Dataset and Features

I used the U.S. Chronic Disease Indicators dataset, published by the Centers for Disease Control and Prevention and retrieved via Data.gov. The dataset contains over 200,000 records across multiple years and includes features like:

- LocationDesc (state)
- Topic (e.g., Alcohol, Cancer)

- Stratification1 (e.g., Age group, Sex)
- DataValue (numerical value of indicator)

The analysis focuses on clustering states based on these indicators to see if any patterns are detected within this data set correlating this to a chronic disease.

Cleaning and Preprocessing

- Removed all nulls in DataValue
- Converted DataValue to numeric type
- Pivoted the data to form a state-by-feature matrix
- Standardized all features using StandardScaler

Modeling Approach

I applied KMeans clustering to the scaled data. The Elbow Method was used to determine the optimal number of clusters.

Tools Used

- Python, Jupyter Notebook
- Libraries: pandas, seaborn, matplotlib, scikit-learn
- Clustering model: KMeans (unsupervised learning)

Results (200 words)

The Elbow Method showed that 4 clusters produced a good tradeoff between inertia and complexity. The final KMeans model grouped the 55 U.S. states/territories into four clusters.

Figure 1 displays the Elbow Method plot used to determine k.
Figure 2 shows a heatmap of alcohol-related indicators across the clusters.

Cluster Distribution:

- Cluster 0: 16 states
- Cluster 1: 20 states
- Cluster 2: 18 states
- Cluster 3: 1 state (high-risk outliers found)

The cluster summary table shows significant variation across demographic groups within the "Alcohol" topic. Cluster 3 showed alcohol-related DataValues as high as 3,154 for certain age groups, compared to less than 100 in other clusters.

This shows that one state had extreme alcohol-related health burdens, while other states were grouped by more moderate or demographic-specific trends.

Discussion (300 words)

The clustering results revealed some patterns in chronic disease. Cluster 3 contained one outlier state with showing alcohol being a concern. Clusters 0 and 2 had the lowest mean values across most indicators, meaning either lower disease prevalence. Cluster 1 showed elevated values, especially in older demographics. One challenge was balancing data coverage and also finding the proper data. Future work would involve including more data like cancer and diabetes. Also, I would like to make it more interactive display so people can easily see my model.

Citations and Attributions

- **Dataset:** Centers for Disease Control and Prevention. U.S. Chronic Disease Indicators. Retrieved from https://chronicdata.cdc.gov/Chronic-Disease-Indicators/U-S-Chronic-Disease-Indicators-CDI-/g4ie-h725

Figures and Captions

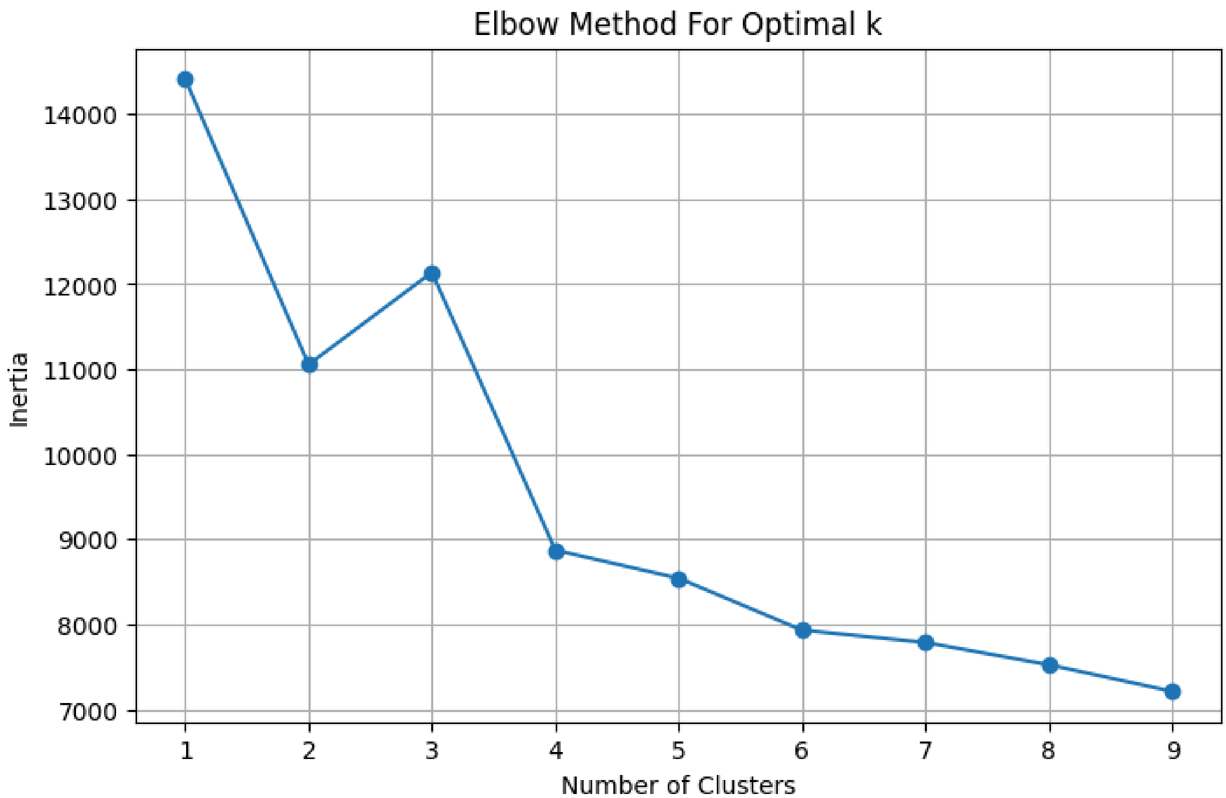

Figure 1. Elbow Method plot showing optimal k = 4 for KMeans clustering.

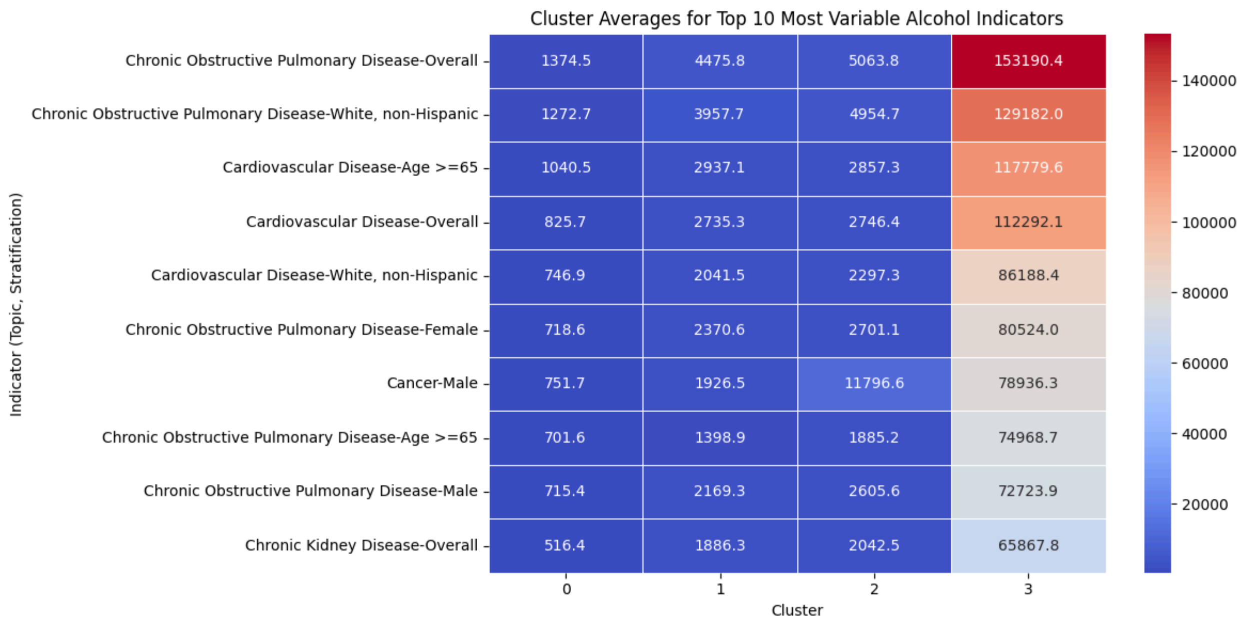

Figure 2. Heatmap of alcohol-related indicator means by cluster, highlighting extreme outlier in Cluster 3.